Ashton Breer

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Education

University of California, Irvine Irvine, CA
Bachelor of Science in Computer Science *Expected Jun. 2028*

Experience

Intern Jun. 2026 – Sept. 2026
PACCAR Leasing *Bellevue, WA*

- Engineered an end-to-end Python and Power Automate workflow to convert incoming email requests into completed Bill of Sale documents, replacing a manual process taking 5–10 minutes per request.
- Developed JSON-based data ingestion and document-processing scripts to parse email threads, decode Base64 attachments, and extract unit numbers for automated vehicle record lookup.
- Built data validation and lookup logic against Excel datasets exported from Tableau and Snowflake, mapping customer, vehicle, serial number, and unit information into standardized Word templates.
- Implemented automated file monitoring, scheduled execution with Windows Task Scheduler, and request-routing logic to streamline recurring processing and reduce repetitive data entry.
- Debugged Python environment, file-handling, and workflow integration issues while transferring the automation to a coworker workstation for continued use after deployment.
- Collaborated with accounting, credit, and insurance teams on leasing operations, including factory invoice reconciliation, change-of-service requests, and document processing in IBS/AS400 and DocuWare.

Style Consultant Summer 2025
Target *Bellevue, WA*

- Supported fast-paced retail operations by resolving customer requests, coordinating with team members, and managing competing service priorities.
- Assisted customers with product selection, merchandise questions, and purchase decisions while maintaining organized displays and supporting team coordination during busy shifts.

Projects

AI Sports Play Prediction Platform | React, FastAPI, Python, scikit-learn Aug. 2025 – Present

- Developed a full-stack React and FastAPI sports analytics application combining machine learning predictions, REST API functionality, and interactive NFL/NBA game visualizations.
- Processed and transformed 356K+ NFL plays and 16M+ NBA events with Python, pandas, and NumPy, preparing structured datasets for predictive modeling and game-state analysis.
- Trained Random Forest and XGBoost models to evaluate historical play outcomes, applying feature engineering and supervised learning techniques to identify patterns in sports data.
- Designed API-driven frontend components and Three.js visualizations to explore model predictions, historical game situations, and interactive analysis of play scenarios.

K-Nearest Neighbors Classifier | Python, NumPy, pandas Feb. 2025 – Mar. 2025

- Implemented a K-Nearest Neighbors (KNN) classifier from scratch, including configurable k-values, Euclidean distance calculations, nearest-neighbor search, and majority-vote classification.
- Optimized numerical operations with NumPy vectorization, reducing distance computation time by 40% while achieving 80% classification accuracy.
- Designed an end-to-end machine learning workflow for data preprocessing, parameter testing, and classification.

Grin Language Interpreter | Python, Object-Oriented Programming May 2025 – Jun. 2025

- Developed a Python interpreter for a custom programming language supporting 13 commands, including variables, conditional branching, loops, and subroutine execution.
- Implemented lexical analysis, parsing, and object-oriented execution logic with support for 6 comparison operators and dynamic program control flow.

Technical Skills

Languages: Python, SQL, C++, C, JavaScript, HTML

Tools & Technologies: Git, GitHub, REST APIs, Power Automate, Microsoft Graph, Snowflake, Salesforce, DocuWare, Power BI, Tableau, Azure, SharePoint, Linux, VS Code, Windows Task Scheduler

AI & Machine Learning: Applied AI, Predictive Modeling, Supervised Learning, Random Forest, XGBoost, Feature Engineering, Model Evaluation, Data Preprocessing

Concepts: Software Engineering, AI-Assisted Development, API Integration, Backend Development, Data Pipelines, ETL, Workflow Automation, Systems Integration, Prototyping, Testing & Debugging, Data Validation, Technical Documentation, Cross-Functional Collaboration